

Gradient Descent

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Gradient Descent pseudocode:

Algorithm 1 Gradient Descent for $f(x) = x^2$

Input: Initial value x_0 , learning rate η , total iterations T

$x \leftarrow x_0$

for $t = 1$ **to** T **do**

$g \leftarrow 2x$ {Compute the gradient $f'(x)$ }

$x \leftarrow x - \eta \cdot g$ {Update the parameter}

$f(x) \leftarrow x^2$ {Compute new function value}

print $t, x, f(x)$

end for

Implement on Python

```
import math

def f(x):
    return x**2

def df(x):
    return 2*x

def gradient_descent(x0, lr, iterations):
    x = x0

    results = []

    for t in range(1, iterations + 1):
        grad = df(x)
        x = x - lr * grad
        fx = f(x)
        results.append([t, x, fx])
        print(f"Time {t}: x = {x:.4f}, f(x) = {fx:.4f}")
```

The result shows that:

Time	x	f(x)
1	8.0000	64.0000
2	6.4000	40.9600
3	5.1200	26.2144
4	4.0960	16.7772
5	3.2768	10.7374
6	2.6214	6.8719
7	2.0972	4.3980
8	1.6777	2.8147
9	1.3422	1.8014
10	1.0737	1.1529

Table 1: Gradient Descent Example